WORKED SOLUTION

STEP II 2008, Question 7

Pure • First Order Differential Equations

STEP II 2008, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/08-s2-q7

The key idea

The left-hand side $\frac{1}{y}\frac{\mathrm{d}y}{\mathrm{d}x}$ is the derivative of $\ln y$, so the substitution is best handled by taking logarithms: a product $y = u\,(1+x^2)^{1/2}$ turns into a sum, and the term $\frac{x}{1+x^2}$ on the right is exactly the derivative of $\ln(1+x^2)^{1/2}$. It cancels, leaving a separable equation for u. Part (ii) is the same idea with the cube root, and once the two answers sit side by side the pattern for (iii) is plain.

Solution

Part (i)

We have $y = 1 \neq 0$ at $x = 0$, and the equation involves $1/y$, so we work on an interval around $x = 0$ on which $y \neq 0$. Since $(1+x^2)^{1/2} > 0$, the substitution gives $u \neq 0$ there too, and $u(0) = 1$.

Taking logarithms of $|y| = |u|\,(1+x^2)^{1/2}$ and differentiating,

$$\frac{1}{y}\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{u}\frac{\mathrm{d}u}{\mathrm{d}x} + \frac{x}{1+x^2}.$$

Comparing with the differential equation, the $\frac{x}{1+x^2}$ terms cancel and we are left with

$$\frac{1}{u}\frac{\mathrm{d}u}{\mathrm{d}x} = xy = x\,u\,(1+x^2)^{1/2}, \qquad \text{so} \qquad \frac{1}{u^2}\frac{\mathrm{d}u}{\mathrm{d}x} = x(1+x^2)^{1/2},$$

dividing by u, which we may do since $u \neq 0$.

Ask yourself How do we integrate $x(1+x^2)^{1/2}$ without a fuss?

The factor x is half the derivative of $1+x^2$, so by the reverse chain rule $\int x(1+x^2)^{1/2}\,\mathrm{d}x = \frac{1}{2}\cdot\frac{2}{3}(1+x^2)^{3/2} = \frac{1}{3}(1+x^2)^{3/2}$, plus a constant. (Differentiate to check.)

Integrating both sides with respect to x,

$$-\frac{1}{u} = \tfrac{1}{3}(1+x^2)^{3/2} + C.$$

At $x = 0$, $u = 1$, so $-1 = \frac{1}{3} + C$ and $C = -\frac{4}{3}$. Hence

$$\frac{1}{u} = \frac{4-(1+x^2)^{3/2}}{3}, \qquad u = \frac{3}{4-(1+x^2)^{3/2}},$$

and therefore

$$\boxed{y = \frac{3\,(1+x^2)^{1/2}}{4-(1+x^2)^{3/2}}}$$

This is valid on the interval containing 0 where the denominator is positive, namely $(1+x^2)^{3/2} < 4$, i.e. $|x| < \sqrt{2^{4/3}-1} \approx 1.233$; as $|x|$ approaches that value, $y \to \infty$.

Part (ii)

Now the extra term is $\frac{x^2}{1+x^3}$, which is the derivative of $\frac{1}{3}\ln(1+x^3) = \ln(1+x^3)^{1/3}$. That tells us to write $y = u\,(1+x^3)^{1/3}$, working near $x = 0$ (where $1+x^3 > 0$). Exactly as before,

$$\frac{1}{y}\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{u}\frac{\mathrm{d}u}{\mathrm{d}x} + \frac{x^2}{1+x^3},$$

so the equation becomes

$$\frac{1}{u}\frac{\mathrm{d}u}{\mathrm{d}x} = x^2 y = x^2 u\,(1+x^3)^{1/3}, \qquad \frac{1}{u^2}\frac{\mathrm{d}u}{\mathrm{d}x} = x^2(1+x^3)^{1/3}.$$

This time x^2 is a third of the derivative of $1+x^3$, so

$$-\frac{1}{u} = \tfrac{1}{3}\cdot\tfrac{3}{4}(1+x^3)^{4/3} + C = \tfrac{1}{4}(1+x^3)^{4/3} + C.$$

With $u = 1$ at $x = 0$, $-1 = \frac{1}{4} + C$, so $C = -\frac{5}{4}$. Thus $\dfrac{1}{u} = \dfrac{5-(1+x^3)^{4/3}}{4}$ and

$$\boxed{y = \frac{4\,(1+x^3)^{1/3}}{5-(1+x^3)^{4/3}}}$$

valid near 0 for $-1 < x < \left(5^{3/4}-1\right)^{1/3} \approx 1.33$.

Part (iii)

Comparing the two answers, the root $(1+x^n)^{1/n}$ appears on top, the power $(1+x^n)^{(n+1)/n}$ appears below, and the constants are $n+1$ and $n+2$. So we conjecture

$$\boxed{y = \frac{(n+1)\,(1+x^n)^{1/n}}{n+2-(1+x^n)^{(n+1)/n}}}$$

which gives the answers to (i) and (ii) when $n = 2$ and $n = 3$. (No proof is asked for, but the same substitution $y = u(1+x^n)^{1/n}$ proves it: $\int x^{n-1}(1+x^n)^{1/n}\,\mathrm{d}x = \frac{1}{n+1}(1+x^n)^{(n+1)/n}$, and the condition at $x = 0$ gives the constant $-\frac{n+2}{n+1}$.)

Checks

- At $x = 0$ each formula gives $\frac{n+1}{n+2-1} = 1$, as required.
- Near $x = 0$ in (i), the equation gives $y'(0) = 0$, and indeed both $(1+x^2)^{1/2}$ and $(1+x^2)^{3/2}$ have zero gradient at 0.
- Solving the equations numerically (a Runge–Kutta step from $x = 0$) matches the formulas to better than 10^{-8} for $n = 2, 3, 4, 5, 7$ at several points. For example in (i) at $x = 1$, $y = \frac{3\sqrt{2}}{4-2\sqrt{2}} \approx 3.6213$, and in (ii) at $x = 1$, $y = \frac{4\cdot 2^{1/3}}{5-2^{4/3}} \approx 2.0320$.

Common slips

- Substituting $y = u(1+x^2)^{1/2}$ and differentiating with the product rule directly: it works, but it is messy. Dividing through by y (equivalently, taking logs) makes the cancellation obvious.
- Forgetting the minus sign in $\int u^{-2}\,\mathrm{d}u = -u^{-1}$, which spoils the constant.
- Fixing the constant at the wrong stage: apply $u(0) = 1$ (not $y(0) = 1$ in the u-equation) before rearranging.

Beyond STEP

The equation is a *Bernoulli equation*: $y' = P(x)y + Q(x)y^2$, and the general trick is the substitution $v = 1/y$, which makes it linear. Here the substitution in the question does that job in two steps: removing the P-term with an integrating-factor-like $(1+x^n)^{1/n}$ and then separating. Notice also that the solution blows up at a finite x even though the equation looks harmless; this "finite-time blow-up" is typical of equations with a y^2 term and is studied in a first course on ODEs. Boyce and DiPrima, *Elementary Differential Equations*, covers Bernoulli equations and the existence of solutions.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.